ACADEMIC CASE STUDY SPECIFICATION

ATM Menu-Driven Banking Simulation

1. PROBLEM STATEMENT

Develop a Python program to simulate ATM operations such as deposit, withdrawal, balance inquiry, and transaction summary with proper validation and account limits.

2. OBJECTIVES

- Deposit money.
- Withdraw money.
- Check account balance.
- Display transaction summary.
- Validate transactions and account limits.

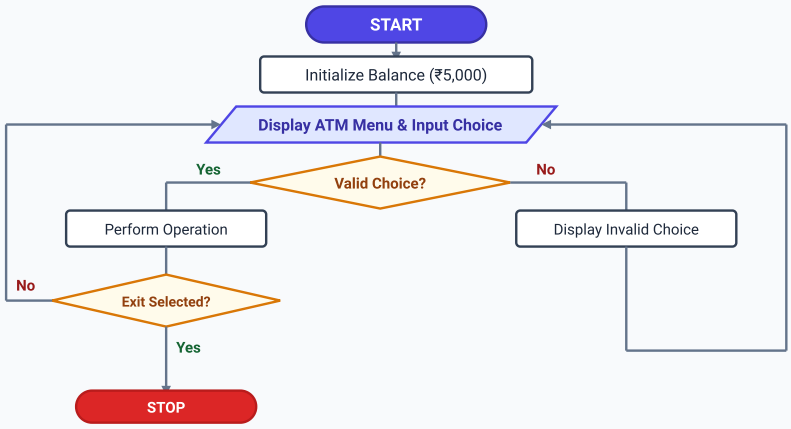
3. ASSUMPTIONS

- Initial balance: ₹5,000.
- Maximum withdrawal: ₹10,000 per transaction.
- Minimum balance: ₹500.
- Deposit limit: ₹20,000 per transaction.
- Invalid amounts are not accepted.

4. ALGORITHM

1. Start.
2. Initialize account balance.
3. Display ATM menu.
4. Select an operation.
5. Perform deposit, withdrawal, or balance inquiry.
6. Validate transaction limits.
7. Display transaction summary when selected.
8. Repeat until Exit.
9. Stop.

5. FLOWCHART



6. PSEUDOCODE

```
START balance = 5000 REPEAT: Display menu Read choice IF choice = 1: Deposit money ELSE IF choice = 2: Withdraw money ELSE IF choice = 3: Display balance ELSE IF choice = 4: Display transactions ELSE IF choice = 5: EXIT ELSE: Display invalid choice STOP
```

7. PYTHON PROGRAM

```
balance = 5000
transactions = []

while True:
    print(""\n1. Deposit"")
    print(""2. Withdraw"")
    print(""3. Balance Inquiry"")
    print(""4. Transaction Summary"")
    print(""5. Exit"")
    choice = input(""Enter choice: "")

    if choice == ""1"":
        amount = float(input(""Enter deposit: ""))
        if 0 < amount <= 20000:
            balance += amount
            transactions.append(""Deposit: Rs."" + str(amount))
            print(""Deposit Successful"")
        else:
            print(""Invalid Deposit"")
    elif choice == ""2"":
        amount = float(input(""Enter withdrawal: ""))
        if 0 < amount <= 10000 and balance - amount >= 500:
            balance -= amount
            transactions.append(""Withdrawal: Rs."" + str(amount))
            print(""Withdrawal Successful"")
        else:
            print(""Invalid Amount or Insufficient Balance"")
    elif choice == ""3"":
        print(""Balance: Rs."", balance)
    elif choice == ""4"":
        print(""Transaction Summary:"")
        for t in transactions:
            print(t)
    elif choice == ""5"":
        print(""Thank You!"")
        break
    else:
        print(""Invalid Choice"")
```

8. SAMPLE OUTPUT

```
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction Summary
5. Exit

Enter choice: 1
Enter deposit: 2000
Deposit Successful

Enter choice: 2
Enter withdrawal: 1000
Withdrawal Successful

Enter choice: 3
Balance: Rs. 6000.0

Enter choice: 4
Transaction Summary:
Deposit: Rs.2000.0
Withdrawal: Rs.1000.0

Enter choice: 5
Thank You!
```

9. TEST CASES

Test #	Operation	Expected Result
1	Deposit ₹2,000	Success
2	Withdraw ₹1,000	Success
3	Withdraw ₹15,000	Invalid
4	Deposit ₹25,000	Invalid
5	Balance inquiry	Display balance
6	Exit	Stop program

10. CONCLUSION

The ATM simulation performs deposits, withdrawals, balance inquiries, and transaction summaries using Python loops, conditions, lists, and input validation.